

Deepak Mallampati

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Kent, OH (open to NYC relocation) - US-based, F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

AI Solutions Engineer with shipped multi-provider AI integrations (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama), FastAPI/Docker service packaging, and a track record of bridging non-technical stakeholders to working AI systems. ~35% retrieval precision gains on healthcare RAG, ~30% post-deploy defect reduction via load simulation, and a Chrome Web Store extension built around real-world API integration across 29 site adapters. Comfortable architecting REST APIs live with stakeholders, owning customer-readable system limitations, and treating AI coding tools as daily colleagues. US-based, F-1 OPT, open to sponsorship.

CORE COMPETENCIES

API Integration • Multi-Provider AI Tooling • RAG / Agentic Workflows • FastAPI / REST • Docker • Stakeholder Documentation • LLM Evaluation • TypeScript / Python • SQL (T-SQL, PostgreSQL)

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - USA (Healthcare Technology, AI Consulting, SaaS)

- Built **Retrieval-Augmented Generation (RAG)** systems for clinical knowledge retrieval and healthcare documentation search; recursive semantic chunking + transformer embeddings improved contextual retrieval precision by ~35% and cut irrelevant retrieval >30%; grounded response alignment exceeded 90%.
- Developed **agentic LLM workflows** for multi-step healthcare queries (eligibility, care navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; raised agent response stability ~25%.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show, care engagement, and support prioritization; improved recall 15-20% on high-risk categories while holding precision >90% via class weighting, stratified sampling, threshold calibration.
- Packaged ML/LLM inference as **FastAPI/Flask** services, containerized with **Docker**, with structured logging and load simulation; reduced post-deploy defects ~30%.
- Built preprocessing pipelines (Pandas, NumPy) for EHR extracts, appointment histories, support ticket logs; raised dataset reliability above 98% and cut downstream model instability ~40%.
- Maintained HIPAA-conscious data governance: de-identification, lineage docs, evaluation audit trails, stakeholder-facing limitation documentation.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Hyderabad, India (Oil & Gas / ERP)

- Optimized SQL and **T-SQL stored procedures** powering the Synthesis Order-to-Cash platform; reduced query execution time ~20% and data retrieval latency ~25%.
- Built **CI/CD pipelines with Jenkins** for schema and stored-procedure deployments; cut deployment errors >30% and release cycle ~35-40%.
- Designed performance dashboards via SQL Server DMVs and **Grafana** for long-running queries, deadlocks, CPU/I/O contention; cut incident recurrence ~25%.
- Tuned database maintenance (index rebuilds, statistics updates, partition-aware indexing); improved reconciliation job runtime ~15%, reduced deadlocks ~15%.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Hyderabad, India (Nonprofit / Video Analytics / Applied NLP)

- Built a **transformer-based video summarization & highlight system** over 5,000+ recorded sessions; hierarchical long-context summarization + timestamp-aligned clip extraction reduced manual review time 60-70%.
- Achieved ~85% **alignment** between AI-selected highlights and human-curated key moments.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); cut hallucinated outputs ~30% and raised contextual accuracy >85% in controlled tests.
- Deployed as a **Flask**-based API with a lightweight UI for non-technical staff.

SELECTED PROJECTS

Manga Lens - Chrome Extension for Real-Time AI Manga/Webtoon Translation shipped, Chrome Web Store

Full-stack browser extension (Manifest V3, content scripts, service workers, captureVisibleTab). Integrates **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v local). Multi-section capture pipeline handles tall webtoon panels via viewport-slice screenshots with coordinate remapping and dedup. Seven-day translation cache, per-domain selector configs for 29 manga/webtoon sites, privacy policy, narrowed host permissions.

Dream Decoder - AI-Powered Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript/Vite/Tailwind app interpreting dreams and synthesizing dreamscapes via coordinated multimodal APIs (Perplexity Sonar for interpretation, Replicate image models for 16:9 visuals). Introduced **intermediate structured prompt transformation layers** - improved contextual alignment ~30% and first-pass image success rate ~25-30% over naïve direct-prompt orchestration.

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake → Validation → Consistency → Duplicate Detection → Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; ANN similarity search for duplicate claim detection; explainable risk scoring with audit-ready reasoning traces.

Agentic Pixel Character Synthesis & Animation Engine ongoing

Phased generative-AI system for identity-consistent, pose-controlled pixel character synthesis with sprite-sheet export. Stable Diffusion fine-tuning + LoRA for identity, ControlNet + OpenPose/MediaPipe for pose conditioning, latent-space consistency for animation smoothness, and an **LLM-based agent orchestrator** decomposing prompts into generation tasks. Backend: PyTorch, Hugging Face Diffusers, FastAPI, Docker.

Video Summarization & Highlight Generation System

Transformer-based hierarchical abstractive summarization mapped back to source timestamps for automated clip extraction. **60-70% reduction** in manual review time, ~85% **highlight alignment** with human curation, Flask-based API for non-technical users.

EDUCATION

M.S., Kent State University - Ohio, USA

Aug 2023 - May 2025

WORK AUTHORIZATION

F-1 OPT - authorized to work in the United States; future sponsorship required.

SKILLS

Forward-Deployed / Solutions: End-to-end LLM/RAG delivery for stakeholders, API integrations, FastAPI/Flask service packaging, Docker, RESTful APIs, observability/logging

LLM & GenAI: RAG, agentic workflows, structured outputs, prompt engineering, evaluation, embeddings, vector search, grounding, guardrails

Languages: Python, TypeScript, React, SQL (T-SQL, PostgreSQL, SQLite), C++ (Arduino), PHP

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, Diffusers, LangChain, LlamaIndex

Integration & Automation: RESTful API integration, AI coding tools (Claude Sonnet, GPT-4o mini, Ollama), Chrome extension automation pipelines, schema-validated JSON contracts

Data: Pandas, NumPy, EHR-style data prep, SQL data modeling

Domains: Healthcare AI, Applied AI, Workflow Automation, Enterprise AI